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20231494

PRGFA3

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Code

```
using System;
using System.Collections.Generic;

// these are the Custom exception for the  feeding errors
5 references
public class FeedingException : Exception
{
    3 references
    public FeedingException(string message) : base(message) { }
}

// Struct for dietary information
10 references
public struct DietInfo
{
    private string foodType;
    private int dailyAmount;

    4 references
    public string FoodType
    {
        get { return foodType; }
        set { foodType = value; }
    }

    1 reference
    public int DailyAmount
    {
        get { return dailyAmount; }
        set
        {
            if (value >= 0)
                dailyAmount = value;
```

```
            throw new ArgumentException("Daily amount must be non-negative.");
        }
    }
}

// Enum for the  animal types
10 references
public enum AnimalType
{
    Mammal,
    Bird,
    Reptile,
    Amphibian,
    Fish

}

// Enum for food types

0 references
public enum FoodType
{
    Meat,
    Vegetation,
    Mixed
}

// Enum for habitat types

11 references
public enum HabitatType
{
    Desert,
```

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```
        Forest,
        Aquatic,
        Arctic,
        Grassland
    }

    // Abstract class for general animal behaviors
    10 references
    public abstract class Animal
    {
        private string name;
        private int age;
        private AnimalType type;
        private DietInfo diet;
        private HabitatType habitat;

        16 references
        public string Name
        {
            get { return name; }
            set { name = value; }
        }

        1 reference
        public int Age
        {
            get { return age; }
            set
            {
                if (value >= 0)
                    age = value;
                else
                    throw new ArgumentException("Age must be non-negative.");
```

```
        }

        1 reference
        public AnimalType Type
        {
            get { return type; }
            set { type = value; }
        }

        4 references
        public DietInfo Diet
        {
            get { return diet; }
            set { diet = value; }
        }

        1 reference
        public HabitatType Habitat
        {
            get { return habitat; }
            set { habitat = value; }
        }

        3 references
        public Animal(string name, int age, AnimalType type, DietInfo diet, HabitatType habitat)
        {
            Name = name;
            Age = age;
            Type = type;
            Diet = diet;
            Habitat = habitat;
        }
```

```
            Habitat = habitat;
        }

        // Abstract methods to be implemented by derived classes
        6 references
        public abstract void Eat(string food);
        4 references
        public abstract void Move();
        4 references
        public abstract void Speak();
    }

    // Interface for feedable animals
    4 references
    public interface IFeedable
    {
        4 references
        void Feed(string food);
    }

    // Interface for movable animals
    3 references
    public interface IMovable
    {
        3 references
        void Move();
    }

    2 references
    public interface IClimbable
    {
        2 references
        void Climb();
```

```
    0 references
    public interface ISwimmable
    {
        0 references
        void Swim();
    }

    0 references
    public interface IFlyable
    {
        0 references
        void Fly();
    }

    // Lion class implementing Animal, IFeedable, and IMovable
    2 references
    public class Lion : Animal, IFeedable, IMovable
    {
        1 reference
        public Lion(string name, int age, AnimalType type, DietInfo diet, HabitatType habitat)
            : base(name, age, type, diet, habitat) { }

        2 references
        public override void Eat(string food)
        {
            if (Diet.FoodType == food)
                Console.WriteLine($"{Name} the Lion is eating {food}.");
            else
                throw new FeedingException($"{Name} the Lion cannot eat {food}.");
        }
```

```
    public override void Move()
    {
        Console.WriteLine($"{Name} the Lion is moving.");
    }

    2 references
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Lion roars.");
    }

    2 references
    public void Feed(string food)
    {
        Eat(food);
    }
}

2 references
public class Rhino : Animal, IFeedable, IMovable
{
    1 reference
    public Rhino(string name, int age, AnimalType type, DietInfo diet, HabitatType habitat)
        : base(name, age, type, diet, habitat) { }

    2 references
    public override void Eat(string food)
    {
        if (Diet.FoodType == food)
            Console.WriteLine($"{Name} the Rhino is eating {food}.");
        else
            throw new FeedingException($"{Name} the Rhino cannot eat {food}.");
```

```
    3 references
    public override void Move()
    {
        Console.WriteLine($"{Name} the Rhino is moving.");
    }

    2 references
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Rhino grunts.");
    }

    2 references
    public void Feed(string food)
    {
        Eat(food);
    }
}

// Koala class implementing Animal, IFeedable, IMovable, and IClimbable
2 references
public class Koala : Animal, IFeedable, IMovable, IClimbable
{
    1 reference
    public Koala(string name, int age, AnimalType type, DietInfo diet, HabitatType habitat)
        : base(name, age, type, diet, habitat) { }

    2 references
    public override void Eat(string food)
    {
        if (Diet.FoodType == food)
            Console.WriteLine($"{Name} the Koala is eating {food}.");
```

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```
        else
            throw new FeedingException($"{Name} the Koala cannot eat {food}.");
    }

    3 references
    public override void Move()
    {
        Console.WriteLine($"{Name} the Koala is moving.");
    }

    2 references
    public override void Speak()
    {
        Console.WriteLine($"{Name} the Koala makes a sound.");
    }

    2 references
    public void Feed(string food)
    {
        Eat(food);
    }

    2 references
    public void Climb()
    {
        Console.WriteLine($"{Name} the Koala is climbing.");
    }
}

0 references
class Program
{
    // List to store animals
    static List<Animal> animals = new List<Animal>();
```

```
    0 references
    static void Main(string[] args)
    {
        Console.WriteLine("Welcome to your Virtual Zoo Management System!");
        bool exit = false;
        while (!exit)
        {
            // Display menu options
            Console.WriteLine("\nPlease select an option:");
            Console.WriteLine("1. Add an animal");
            Console.WriteLine("2. Interact with an animal");
            Console.WriteLine("3. Exit");
            Console.Write("Enter your choice: ");
            int choice;
            if (int.TryParse(Console.ReadLine(), out choice))
            {
                switch (choice)
                {
                    case 1:
                        try
                        {
                            AddAnAnimal();
                        }
                        catch (ArgumentException ex)
                        {
                            Console.WriteLine(ex.Message);
                        }
                        break;
                    case 2:
                        InteractWithAnAnimal();
                        break;
                    case 3:
```

```
                    InteractWithAnAnimal();
                    break;
                case 3:
                    exit = true;
                    break;
                default:
                    Console.WriteLine("Invalid choice. Please try again.");
                    break;
                }
            }
            else
            {
                Console.WriteLine("Invalid input. Please enter a number.");
            }
        }
    }

    // Method to add an animal
    1 reference
    static void AddAnAnimal()
    {
        Console.WriteLine("\nChoose an animal to add:");
        Console.WriteLine("1. Lion");
        Console.WriteLine("2. Rhino");
        Console.WriteLine("3. Koala");
        Console.Write("Enter your choice: ");
        int choice;
        if (int.TryParse(Console.ReadLine(), out choice))
        {
            Console.Write("Enter animal's name: ");
            string name = Console.ReadLine();
            Console.Write("Enter animal's age: ");
            int age;
```

```
            if (int.TryParse(Console.ReadLine(), out age))
            {
                // Get animal type
                Console.WriteLine("Enter animal type:");
                foreach (var type in Enum.GetValues(typeof(AnimalType)))
                {
                    Console.WriteLine($"{(int)type}. {type}");
                }
                int typeChoice;
                if (int.TryParse(Console.ReadLine(), out typeChoice) && Enum.IsDefined(typeof(AnimalType), typeChoice))
                {
                    AnimalType animalType = (AnimalType)typeChoice;

                    // Get diet information
                    Console.WriteLine("Enter diet information:(Meat,Vegetation,Mixed) ");
                    DietInfo dietInfo = GetDietInfo();

                    // Get habitat type
                    Console.WriteLine("Enter habitat type:");
                    HabitatType habitatType = GetHabitatType();

                    // Add animal based on choice
                    switch (choice)
                    {
                        case 1:
                            animals.Add(new Lion(name, age, animalType, dietInfo, habitatType));
                            Console.WriteLine("Lion added successfully!");
                            break;
                        case 2:
                            animals.Add(new Rhino(name, age, animalType, dietInfo, habitatType));
                            Console.WriteLine("Rhino added successfully!");
                            break;
                        case 3:
```

```
                    animals.Add(new Koala(name, age, animalType, dietInfo, habitatType));
                    Console.WriteLine("Koala added successfully!");
                    break;
                default:
                    Console.WriteLine("Invalid choice.");
                    break;
                }
            }
            else
            {
                Console.WriteLine("Invalid animal type.");
            }
        }
        else
        {
            Console.WriteLine("Invalid age. Please enter a number.");
        }
    }
    else
    {
        Console.WriteLine("Invalid input. Please enter a number.");
    }
}

// Method to get diet information
1 reference
static DietInfo GetDietInfo()
{
    DietInfo dietInfo = new DietInfo();

    Console.Write("Enter food type: ");
    dietInfo.FoodType = Console.ReadLine();
```

```
    Console.Write("Enter food type: ");
    dietInfo.FoodType = Console.ReadLine();

    Console.Write("Enter daily amount (in grams): ");
    int dailyAmount;
    if (int.TryParse(Console.ReadLine(), out dailyAmount))
    {
        dietInfo.DailyAmount = dailyAmount;
    }
    else
    {
        throw new ArgumentException("Invalid daily amount. Please enter a number.");
    }

    return dietInfo;
}

// Method to get habitat type
1 reference
static HabitatType GetHabitatType()
{
    foreach (var habitat in Enum.GetValues(typeof(HabitatType)))
    {
        Console.WriteLine($"{(int)habitat}. {habitat}");
    }
    int habitatChoice;
    if (int.TryParse(Console.ReadLine(), out habitatChoice) && Enum.IsDefined(typeof(HabitatType), habitatChoice))
    {
        return (HabitatType)habitatChoice;
    }
    else
    {
        throw new ArgumentException("Invalid habitat type.");
```

```
        else
        {
            throw new ArgumentException("Invalid habitat type.");
        }
    }

    // Method to interact with an animal
    1 reference
    static void InteractWithAnAnimal()
    {
        if (animals.Count == 0)
        {
            Console.WriteLine("No animals available to interact with.");
            return;
        }

        Console.WriteLine("\nChoose an animal to interact with:");
        for (int i = 0; i < animals.Count; i++)
        {
            Console.WriteLine($"{i + 1}. {animals[i].Name} the {animals[i].GetType().Name}");
        }

        int choice;
        if (int.TryParse(Console.ReadLine(), out choice) && choice > 0 && choice <= animals.Count)
        {
            Animal animal = animals[choice - 1];

            Console.WriteLine($"\nInteracting with {animal.Name} the {animal.GetType().Name}:");
            animal.Move();
            animal.Speak();

            if (animal is IFeedable feedable)
            {
```

```
            if (animal is IFeedable feedable)
            {
                Console.Write("Enter the food to feed it: ");
                string food = Console.ReadLine();
                try
                {
                    feedable.Feed(food);
                }
                catch (FeedingException ex)
                {
                    Console.WriteLine(ex.Message);
                }
            }

            if (animal is IClimbable climbable)
            {
                climbable.Climb();
            }
        }
        else
        {
            Console.WriteLine("Invalid choice.");
        }
    }
}
```

Output

```
C:\Users\OEM\Desktop\forma

Welcome to your Virtual Zoo Management System!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice:
```

```
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Welcome to your Virtual Zoo Management System!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age:
```

```
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Welcome to your Virtual Zoo Management System!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
```

```
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Welcome to your Virtual Zoo Management System!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type:
```

```
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Welcome to your Virtual Zoo Management System!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams):
```

```
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Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams): 1500
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
```

```
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Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams): 1500
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
```

```
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1. Lion
2. Rhino
3. Koala
Enter your choice: 1
Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams): 1500
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
4
Lion added successfully!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice:
```

```
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Enter animal's name: Tommy
Enter animal's age: 3
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams): 1500
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
4
Lion added successfully!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
```

```
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2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams): 1500
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
4
Lion added successfully!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it:
```

```
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2. Reptile
3. Amphibian
4. Fish
0
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: meat
Enter daily amount (in grams): 1500
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
4
Lion added successfully!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it: mixed
```

```
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Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
4
Lion added successfully!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it: mixed
Tommy the Lion cannot eat mixed.

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice:
```

```
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Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it: mixed
Tommy the Lion cannot eat mixed.

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it:
```

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```
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1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it: Meat
Tommy the Lion cannot eat Meat.

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
1

Interacting with Tommy the Lion:
Tommy the Lion is moving.
Tommy the Lion roars.
Enter the food to feed it: meat
Tommy the Lion is eating meat.

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice:
```

```
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Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 1

Choose an animal to add:
1. Lion
2. Rhino
3. Koala
Enter your choice: 2
Enter animal's name: 12
Enter animal's age: 13
Enter animal type:
0. Mammal
1. Bird
2. Reptile
3. Amphibian
4. Fish
2
Enter diet information:(Meat,Vegetation,Mixed
Enter food type: mixed
Enter daily amount (in grams): 123
Enter habitat type:
0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
```

```
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0. Desert
1. Forest
2. Aquatic
3. Arctic
4. Grassland
2
Rhino added successfully!

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice: 2

Choose an animal to interact with:
1. Tommy the Lion
2. 12 the Rhino
2

Interacting with 12 the Rhino:
12 the Rhino is moving.
12 the Rhino grunts.
Enter the food to feed it: mixed
12 the Rhino is eating mixed.

Please select an option:
1. Add an animal
2. Interact with an animal
3. Exit
Enter your choice:
```